CAT403 COMPUTING INFRASTRUCTURE MAJOR PROJECT

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COGNISHIELD

A Large-Language-Assisted Campus Network Orchestration and Automation Platform

Presented By:

Joshua Lim Miin Ning

limmiinning@gmail.com

Supervised By:

Dr Vaithegy A/P Doraisamy

vaithegy@usm.my

INTRODUCTION

COGNISHIELD

A Large-Language-Assisted Campus Network Orchestration and Automation Platform

“CogniShield is an open-source, vendor-neutral network operations platform that supports LLM-assisted configuration workflows, topology visibility, monitoring, backup, and approval-governed network automation.”

Introduction

Cognishield *is an infrastructure-backed, controller-style platform for network operations supported by automation and AI*

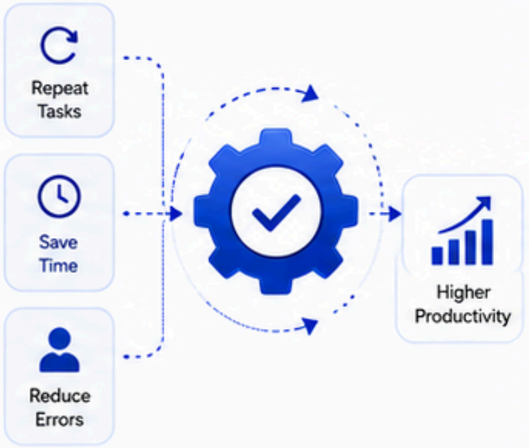
01

PROBLEM

Network operations are still fragmented across CLI, spreadsheets, dashboards, and scripts.

02

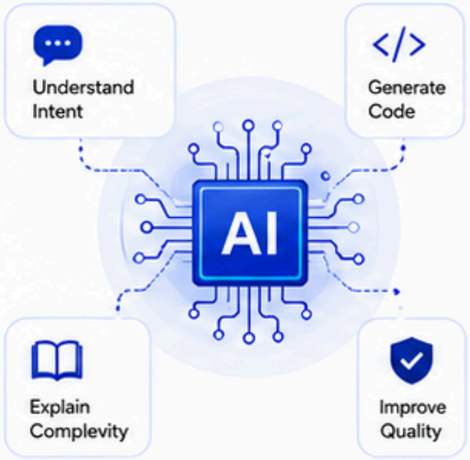
EXISTING SOLUTION



Network automation helps reduce repetitive manual configuration work, improving efficiency and accuracy.

03

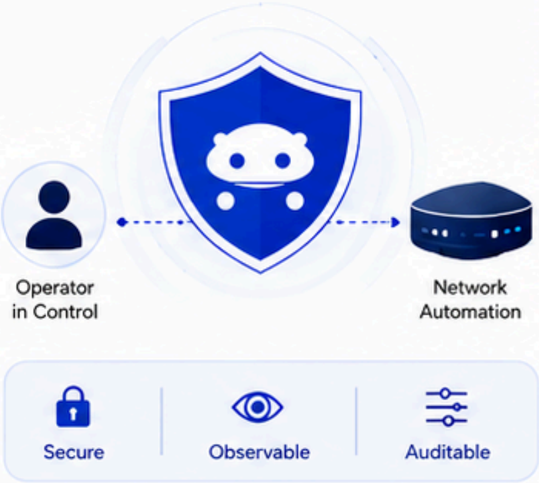
INDUSTRY TREND



AI assistants are transforming software engineering workflows, boosting speed and code quality.

04

CONTRIBUTION



CogniShield brings LLM assistance into network automation while keeping operators in control.

Faster operations. • Smarter decisions. • Stronger networks. **All with CogniShield**

PROBLEM STATEMENTS

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- **Expensive & Proprietary Controller Platforms**
- **Manual, Repetitive, Error-Prone Multi-Vendor Configuration Work**
- **Lack of Governed AI-Assisted Network Automation Workflow**

SYSTEM OBJECTIVES

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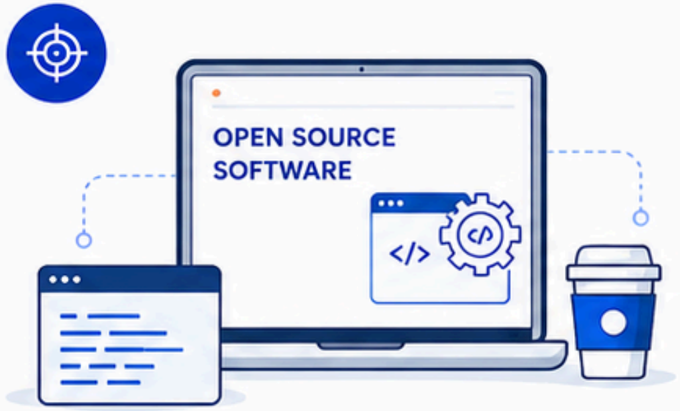
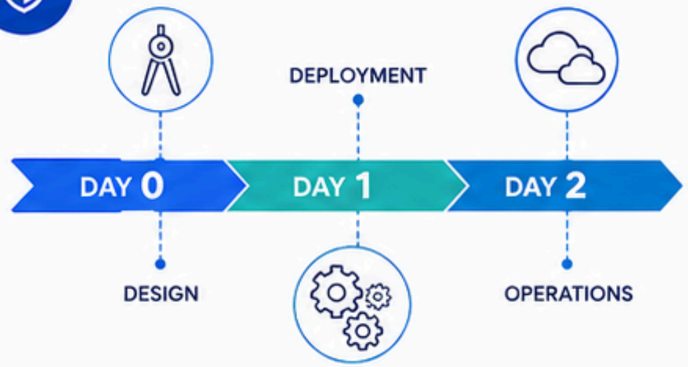


01

To Develop an Accessible, Open-Source Network Management Platform

02

To Implement an LLM-Assisted Validated Automation Workflow for Multi-Vendor Networks



03

To Build an Integrated Governed Day-1 and Day-2 Operations Framework

PROPOSED SOLUTION

PROPOSED SOLUTION

CogniShield — the unified platform for modern, intelligent and governed network operations.

01

Centralized Controller-Style Workspace

CogniShield unifies network operations into one web-based platform.

Inventory | Topology | Monitoring | Backup

02

Unified Operational Modules

Inventory, topology, onboarding, monitoring, backup, AI assistance, and Ansible automation.

CogniShield

Unified • Intelligent • Governed Network Operations

03

Governed Network Automation

Intent Parsing → Ansible Playbook Generation → Dry-Run Simulation → Approval Workflow → Deployment → Verification

Converts operator requests into Ansible workflows with dry-run, approval, deployment, and verification.

04

Local and Secure Deployment

Locally hosted, VPN-accessible, and deployable on practical infrastructure.

Unified Operations | Governed Automation | Secure by Design | Better Visibility | Operational Efficiency | Built for Network Operators

SYSTEM MODULE

MODULE DIAGRAM

COGNISHIELD SYSTEM

NETWORK TOPOLOGY VISUALIZATION MODULE

Purpose:

Visualize and compare planned and live network topologies.

Features:

- Planned Topology View
- Live Topology View
- Topology Comparison (Diff)
- Device & Connection Details

LLM NETWORK AUTOMATION MODULE

Purpose:

Request Parsing, Ansible playbook generation, dry-run, approval, deployment, verification.

Features:

- Request Parsing
- Ansible Playbook Generation
- Dry-Run Simulation
- Approval Workflow
- Deployment & Verification

OPERATIONS SUPPORT MODULE

Purpose:

Streamline day-to-day operations with tools for onboarding, troubleshooting, backup, and AI assistance.

Features:

- Onboarding
- Templates
- Service Profiles
- SSH Terminal
- Syslog, Backup & Diff, AI Assistant

PERFORMANCE METRIC MONITORING MODULE

Purpose:

Monitor network performance and system health.

Features:

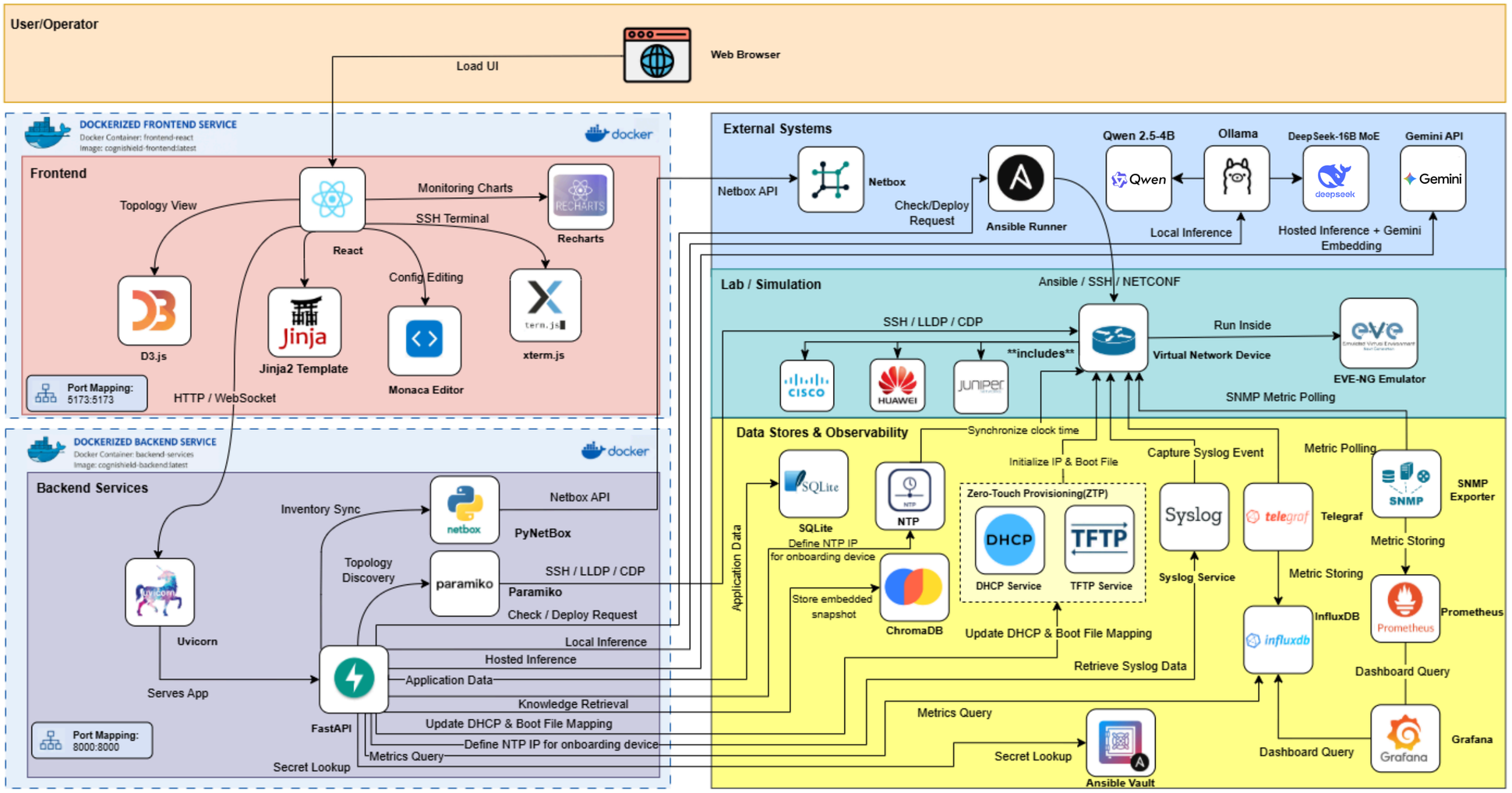
- Performance Metrics
- Traffic Statistics
- Alerts & Notifications
- Trend Analysis & Reports
- System Health Overview

SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

PROXMOX

The Proxmox environment is used as the virtualization layer to host and isolate the different modules and services of the system. Each major component is deployed in its own virtual machine, making the lab setup easier to manage, scale, and recover while simulating a real-world deployment.



IMPACT & BENEFITS

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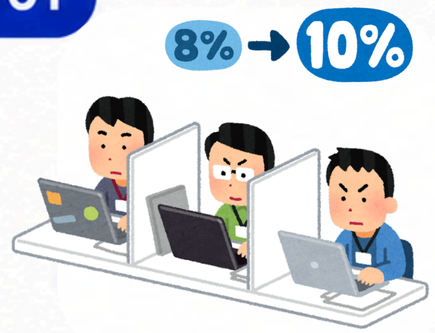
CogniShield delivers measurable value across teams, institutions, businesses, and learners.

01



Network Teams

Faster configuration deployment, fewer manual errors, better multi-vendor consistency.

02

Universities

Practical learning environment for network automation and AI-assisted operations.

03

SMEs

Entry-level automation platform without expensive controller licensing.

04

Students

Hands-on exposure to intent parsing, Ansible automation, topology, monitoring, and verification.

Making AI-assisted network automation more accessible, practical, and governable.

INNOVATION

INNOVATION

CogniShield brings a new paradigm to network automation by combining openness, intelligence, and governance.

01

Open-Source Controller-Style Platform

Network automation platform built around accessible and open tooling.

02

Local LLM Integration

LLM support for interpreting network operation requests.

03

Governed Automation Workflow

From operator request to verified device change.

CogniShield

Unified • Intelligent • Governed Network Automation

04

Vendor-Neutral Architecture

Designed for mixed-vendor network environments.

05

Affordable Local Deployment

Runs on practical local infrastructure.

Not just AI generation – governed AI-assisted network automation.

ENTREPRENEURSHIP OPPORTUNITIES

ENTREPRENEURSHIP OPPORTUNITY

CogniShield
Intelligent • Open • Governed

CogniShield has strong potential to create impact and commercial value across education, businesses, and service providers.

01

TARGET CUSTOMERS

- Universities
- Polytechnics
- SMEs
- MSPs

02

BUSINESS MODEL

Community	Free
Professional	Subscription
Enterprise	Support Contract
Academic	Campus Licensing

03

REVENUE STREAMS

- Deployment services
- Training workshops
- Support contracts
- Enterprise features

Affordable, vendor-neutral network operations tooling for education, SMEs, and service providers.

SYSTEM DEMO WORKFLOW

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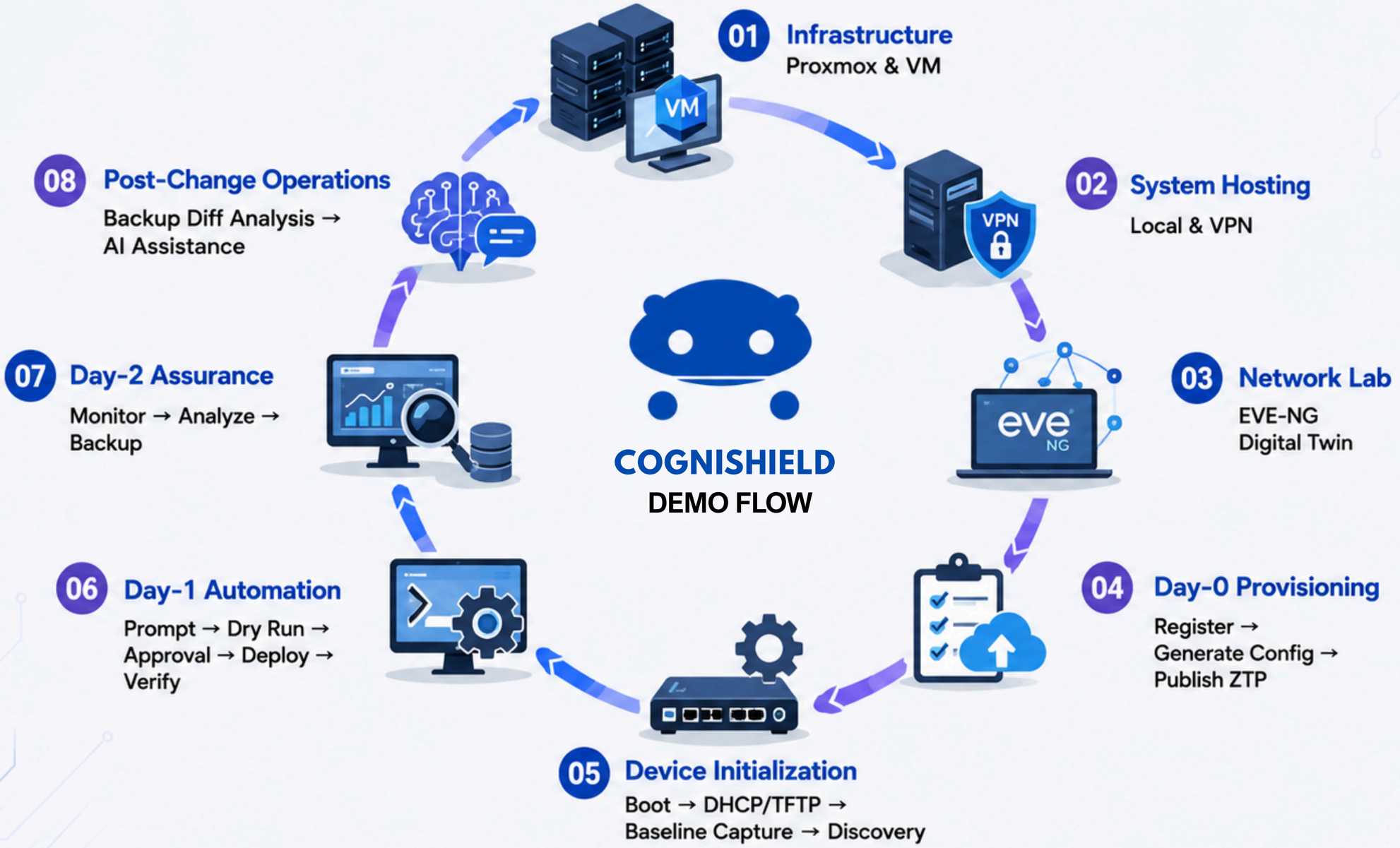
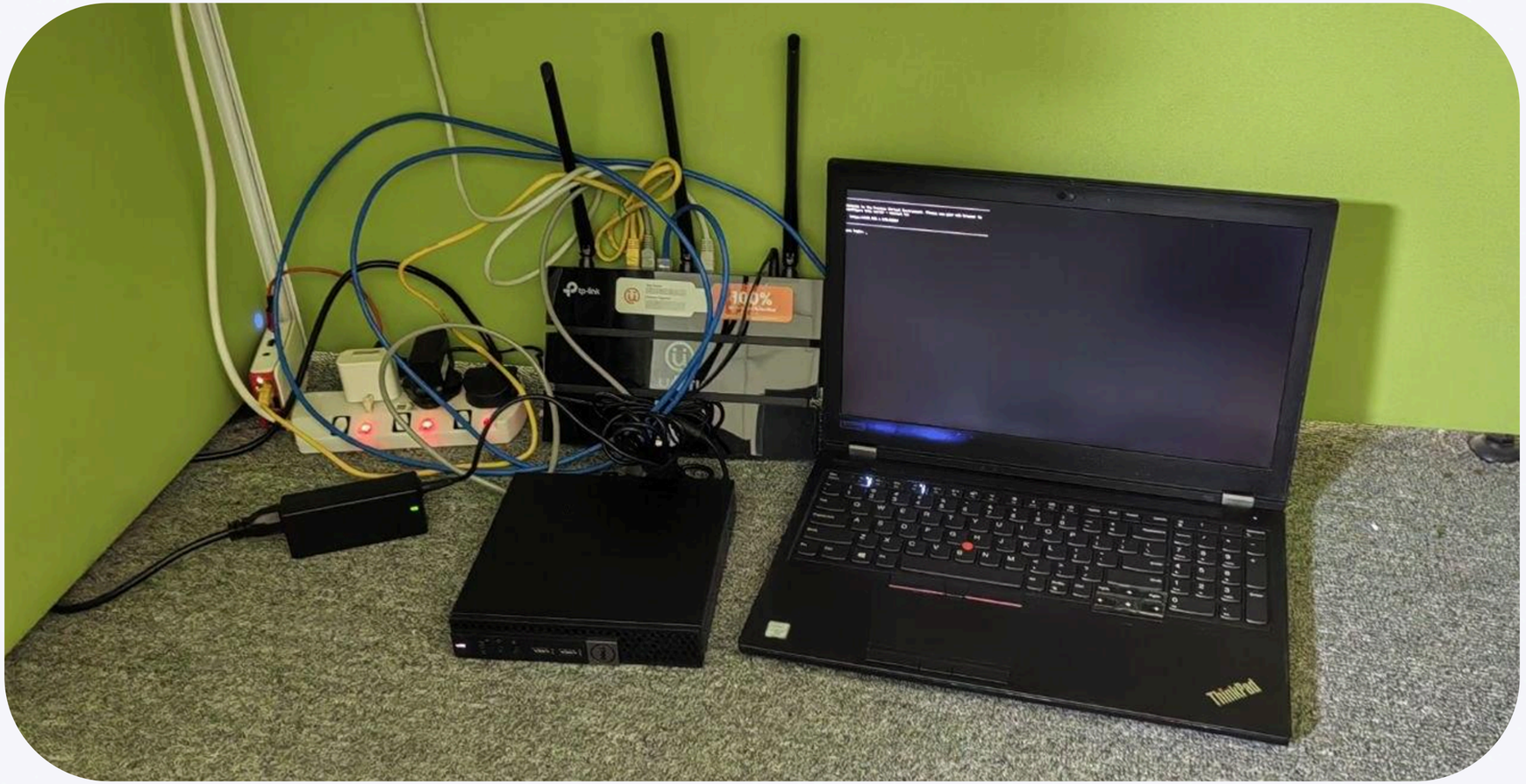


DEMO TIME

PROXMOX SERVER & LOCAL NETWORK SETUP



CONCLUSION

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1. CogniShield successfully achieved a **functional MVP-oriented** network operations platform.
2. The project unified fragmented **topology visibility**, **inventory context**, **automation workflow**, and **monitoring support**.
3. It demonstrated **governed AI-assisted operations** rather than unrestricted autonomous control.
4. It proved the **feasibility** of building a controller-style platform with open-source tools and accessible local infrastructure.
5. It provides a **meaningful foundation** for future strengthening and extension.

THANK YOU !!